

Executive Business Report

End-to-End Sales Forecasting & Demand Intelligence System

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Executive Summary

This project was developed to build an intelligent sales forecasting and demand analysis system for retail businesses using historical Superstore sales data. The objective was to forecast future sales, identify unusual sales patterns, segment products based on demand behaviour, and present business insights through an interactive Streamlit dashboard. Multiple forecasting models, including SARIMA, Facebook Prophet, and XGBoost, were implemented and compared. The project provides valuable insights that can help businesses improve inventory planning, reduce stock shortages, and support data-driven decision-making.

Key Findings from Exploratory Data Analysis

The exploratory analysis revealed several important business insights:

- The **Technology** category generated the highest overall sales among all product categories.
- The **West** region recorded the highest total sales compared to the other regions.
- Monthly sales showed an overall upward trend with noticeable seasonal fluctuations across the four-year period.
- The **Consumer** customer segment contributed the largest share of total sales.

- The average shipping time was approximately **3.96 days**, indicating efficient order fulfilment.

Sales Forecasting Results

Three forecasting models were implemented and evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE).

Model	MAE	RMSE	MAPE
SARIMA	18031.40	19009.18	18.97%
Facebook Prophet	20250.79	22318.41	21.86%
XGBoost	15906.45	17771.28	23.05%

Among the three forecasting models, **SARIMA** achieved the lowest MAPE (18.97%) and was therefore selected as the best-performing forecasting model for this dataset.

Anomaly Detection

Isolation Forest was applied to detect unusual sales transactions. The model successfully identified abnormal sales records that may represent exceptionally high-value orders, promotional events, seasonal demand spikes, or data inconsistencies requiring further investigation. Early detection of such anomalies can help businesses improve operational monitoring and decision-making.

Product Demand Segmentation

K-Means clustering was used to group products based on their sales characteristics. The algorithm divided the products into three clusters representing different demand patterns. This segmentation enables businesses to identify products with similar demand behaviour and supports better inventory planning and stock management decisions.

Business Recommendations

Based on the analysis, the following recommendations are proposed:

1. Maintain Higher Inventory for Technology Products

Technology products generated the highest sales and should receive higher inventory priority to minimise stock-out situations.

2. Improve Seasonal Inventory Planning

Historical sales patterns indicate recurring seasonal fluctuations. Businesses should prepare inventory in advance for high-demand periods to reduce shortages and improve customer satisfaction.

3. Monitor Anomalous Sales Regularly

Implementing automated anomaly detection can help businesses quickly identify unusual sales behaviour, enabling faster investigation and more informed business decisions.

Limitations

The forecasting models were trained using approximately four years of historical sales data. Although the models provide valuable insights, forecasting accuracy may be affected by future market changes, economic conditions, promotional campaigns, and other external factors that are not represented in the historical dataset.

Conclusion

This project successfully demonstrates an end-to-end sales forecasting pipeline, including data preprocessing, exploratory data analysis, time-series forecasting, anomaly detection, product demand segmentation, and deployment through a Streamlit dashboard. The developed system provides meaningful business insights that can support inventory planning, improve operational efficiency, and assist retail organisations in making data-driven decisions.